Pipeline ETL de streaming con Kafka y ML: predicciones en tiempo real del World Happiness Score

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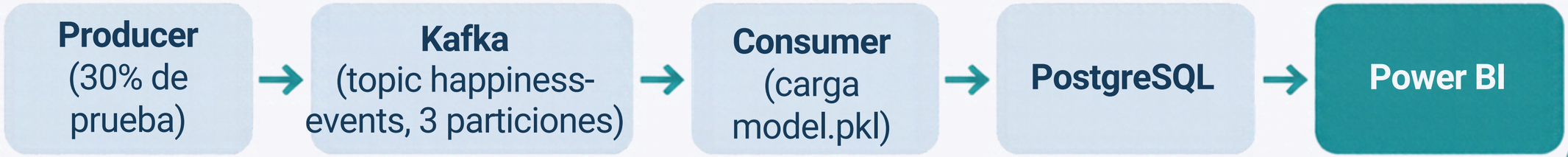
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Universidad Autónoma de Occidente

Arquitectura

Offline



Streaming



El reto: 5 archivos, esquemas distintos

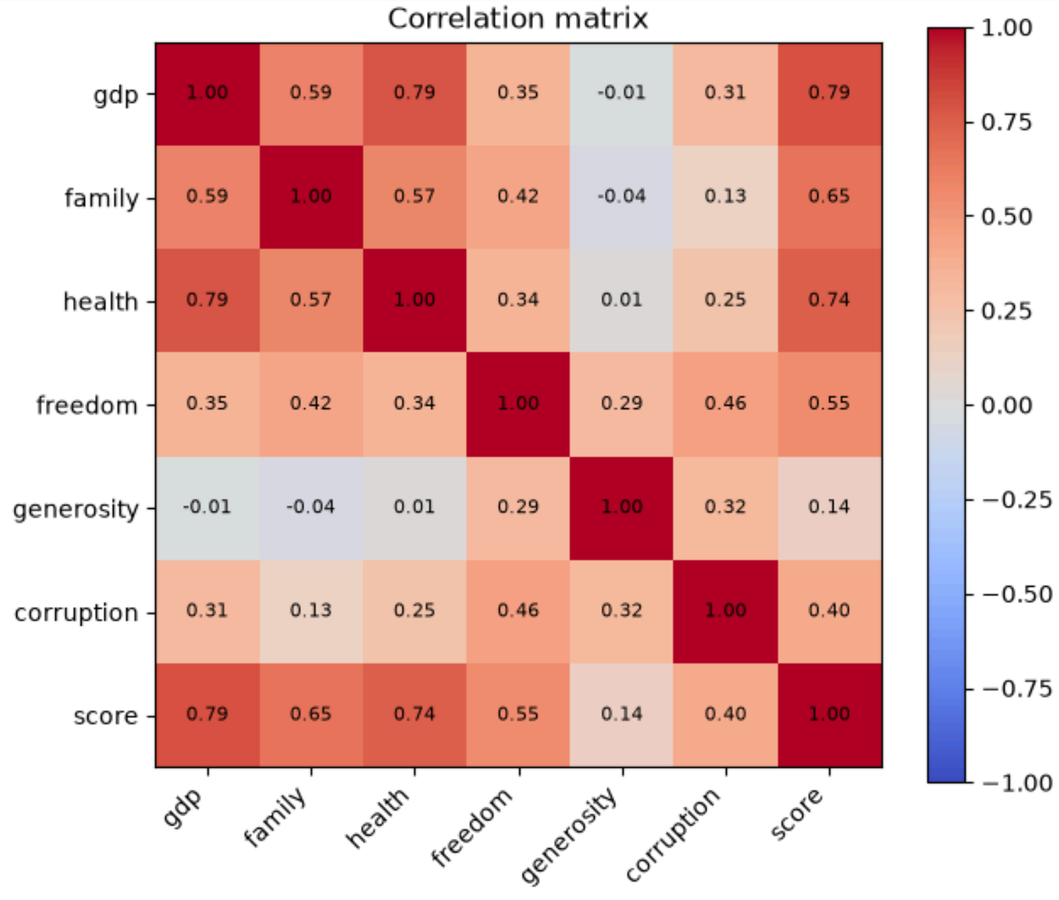
- 782 filas y 12/13/12/9/9 columnas según el año.
- El mismo indicador con 3 nombres: unificados en 2 etapas (limpieza y mapeo)
- 6 países con dos nombres (170 → 164); 1 fila eliminada (Emiratos 2018).
- Residual de distopía excludido: puntaje = indicadores + residual (diferencia 0.0005)

case	variant (replaced)	years of variant	kept name	years of kept name
1	Hong Kong S.A.R., China	2017	Hong Kong	2015, 2016, 2018, 2019
2	Taiwan Province of China	2017	Taiwan	2015, 2016, 2018, 2019
3	North Cyprus	2015, 2016, 2017	Northern Cyprus	2018, 2019
4	Macedonia	2015, 2016, 2017, 2018	North Macedonia	2019
5	Trinidad & Tobago	2018, 2019	Trinidad and Tobago	2015, 2016, 2017
6	Somaliland region	2015	Somaliland Region	2016

year	max |score - (six indicators + dystopia_residual)|
2015	0.0005
2016	0.0005
2017	0.0005

Características y modelo

- 6 indicadores como características; country y year son identificadores.
- **Pipeline =** StandardScaler + LinearRegression.
- **Métricas en el 30% de prueba:** MAE 0.4321 · RMSE 0.5566 · R² 0.7519



	train	test
MAE	0.4225	0.4321
RMSE	0.5448	0.5566
R2	0.7675	0.7519

	coefficient (standardized features)
gdp	0.431
health	0.267
freedom	0.249
family	0.210
corruption	0.099
generosity	0.068

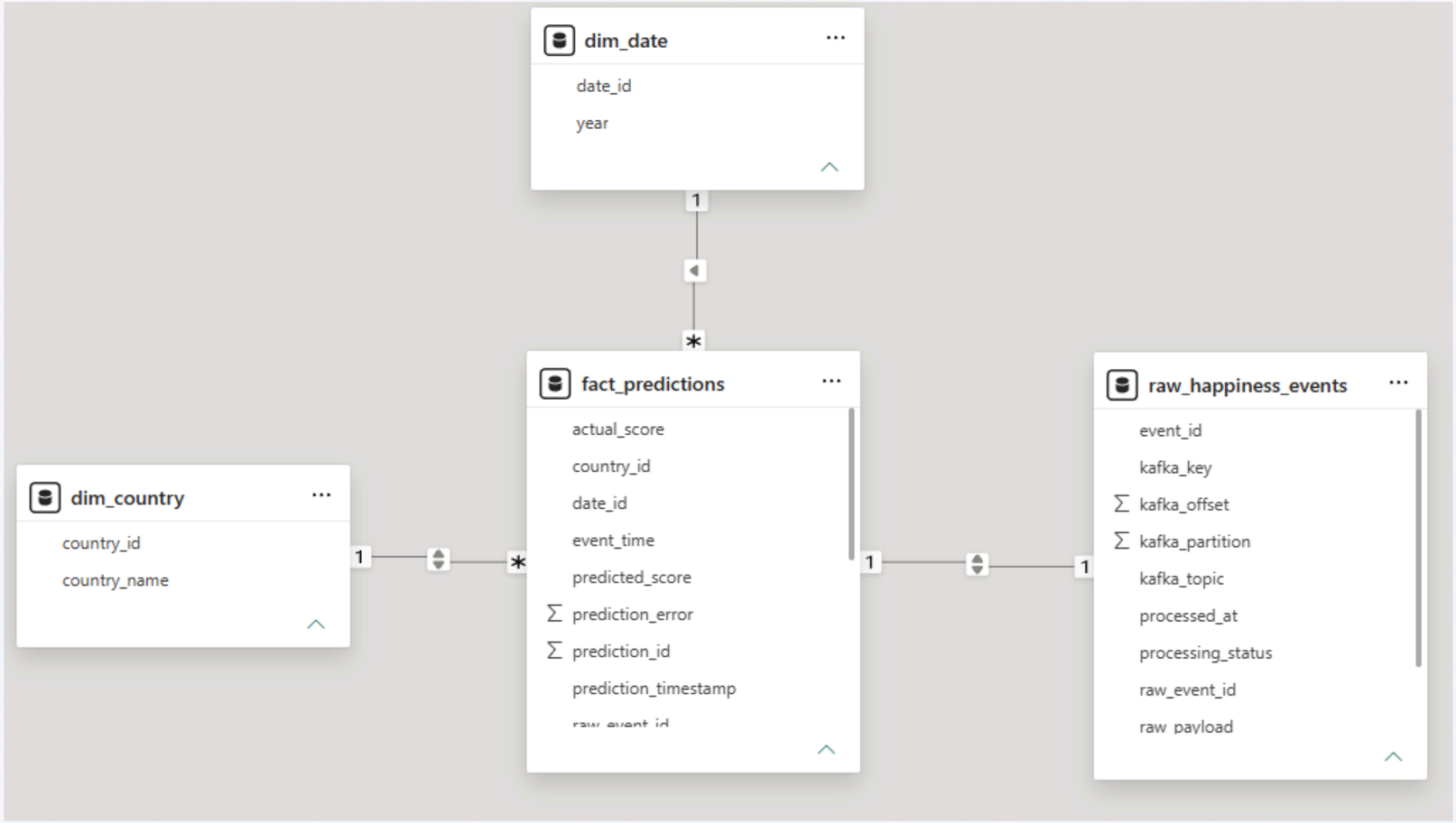
Kafka y consumer

- **Clave =** country; contrato de eventos de 11 campos.
- **4 estados:** VALID, INVALID_SCHEMA, INVALID_VALUES, PREDICTION_ERROR.
- **Guardar crudo → validar → predecir → guardar.**

```
--- 225 predictions so far | running mean absolute error 0.4333
PROCESSED | E226 | Switzerland 2019 | actual 7.480 | predicted 7.208 | error +0.272 | par
PROCESSED | E227 | Syria 2019 | actual 3.462 | predicted 3.888 | error -0.426 | partition
PROCESSED | E228 | Taiwan 2019 | actual 6.446 | predicted 6.305 | error +0.141 | partitio
PROCESSED | E229 | Tanzania 2019 | actual 3.231 | predicted 4.745 | error -1.514 | partit
PROCESSED | E230 | Turkey 2019 | actual 5.373 | predicted 5.619 | error -0.246 | partitio
PROCESSED | E231 | United Arab Emirates 2019 | actual 6.825 | predicted 6.771 | error +0.
PROCESSED | E232 | United Kingdom 2019 | actual 7.054 | predicted 6.817 | error +0.237 |
PROCESSED | E233 | Uruguay 2019 | actual 6.293 | predicted 6.307 | error -0.014 | partiti
PROCESSED | E234 | Venezuela 2019 | actual 4.707 | predicted 5.286 | error -0.579 | parti
PROCESSED | E235 | Vietnam 2019 | actual 5.175 | predicted 5.750 | error -0.575 | partiti
REJECTED | X001 | INVALID_SCHEMA | missing field: gdp | partition 0 offset 88
REJECTED | X002 | INVALID_SCHEMA | freedom must be a number | partition 2 offset 81
REJECTED | X003 | INVALID_VALUES | corruption must be >= 0: -0.5 | partition 0 offset 89
REJECTED | X004 | PREDICTION_ERROR | predicted score out of range [0, 10]: 106599.8 | par
REJECTED | E001 | INVALID_VALUES | duplicate event_id: E001 | partition 1 offset 67
```

Base de datos

- Payload crudo en TEXT; estado RECEIVED.
- Trazabilidad con raw_event_id.

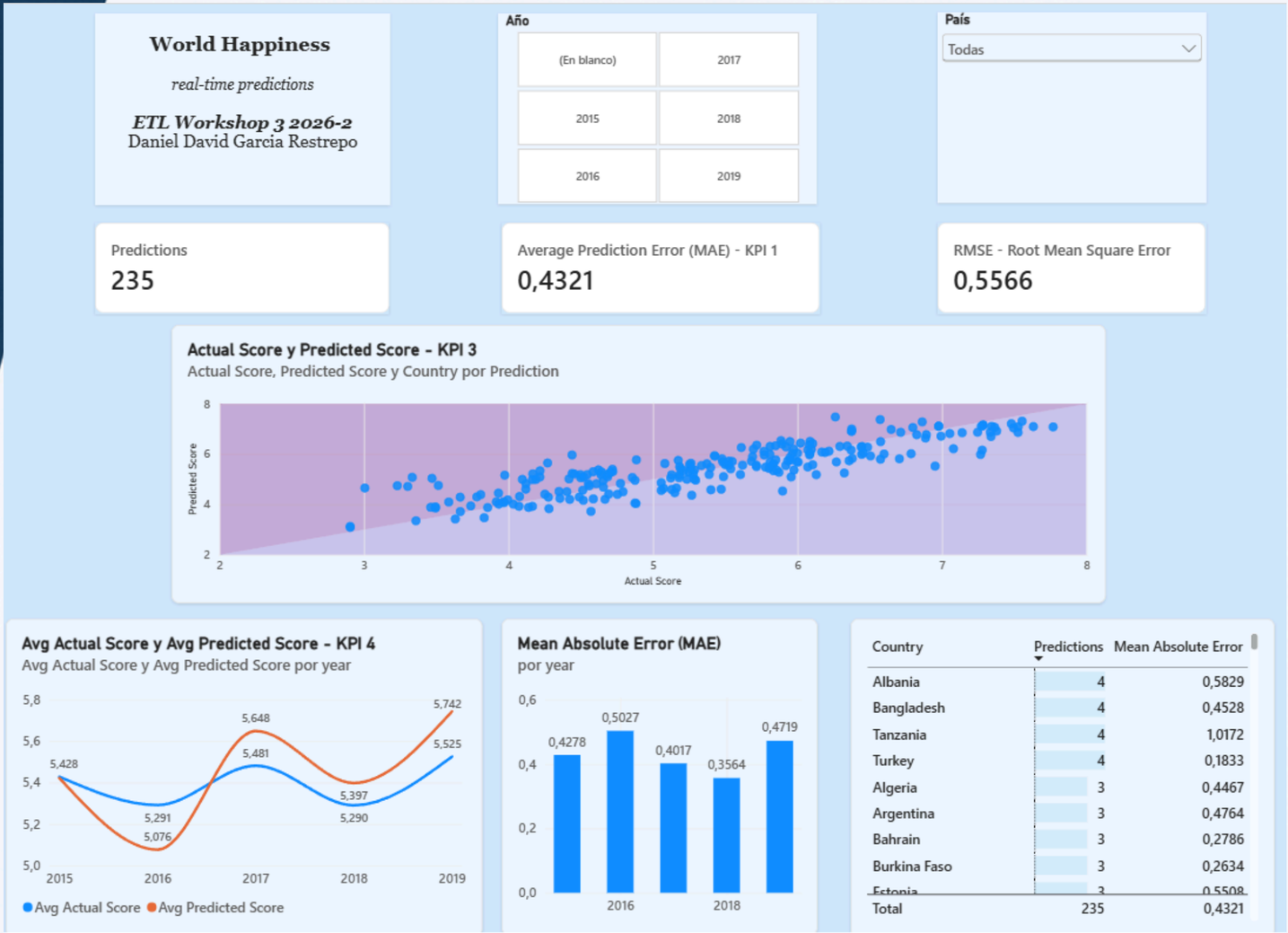


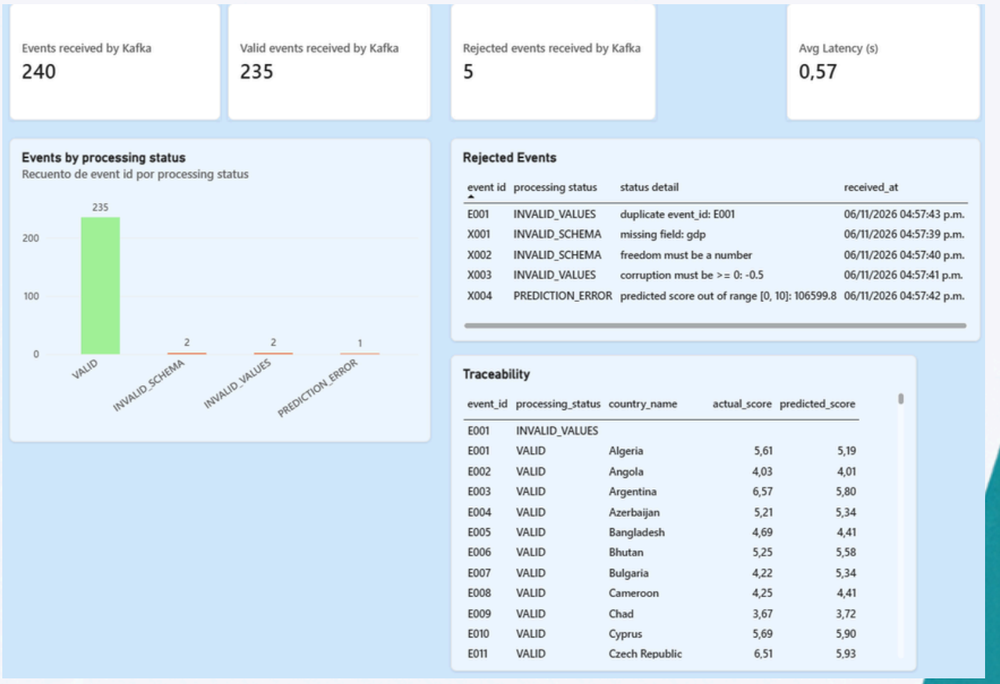
Demo en vivo

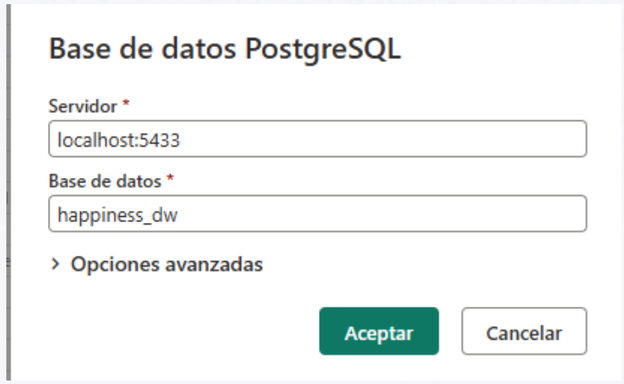
1. Consumer escuchando.
2. Stream limpio (235 eventos).
3. Cinco registros defectuosos.
4. Resultados en SQL.
5. Dashboard.

Dashboard

- **4 KPIs:** error promedio, predicciones por país, predicho vs real y tendencia por año







Resultados y limitaciones

Resultados:

235 predicciones; MAE 0.4321 y RMSE 0.5566.

Limitaciones:

partición aleatoria por fila, panel desbalanceado y broker local sin volumen.

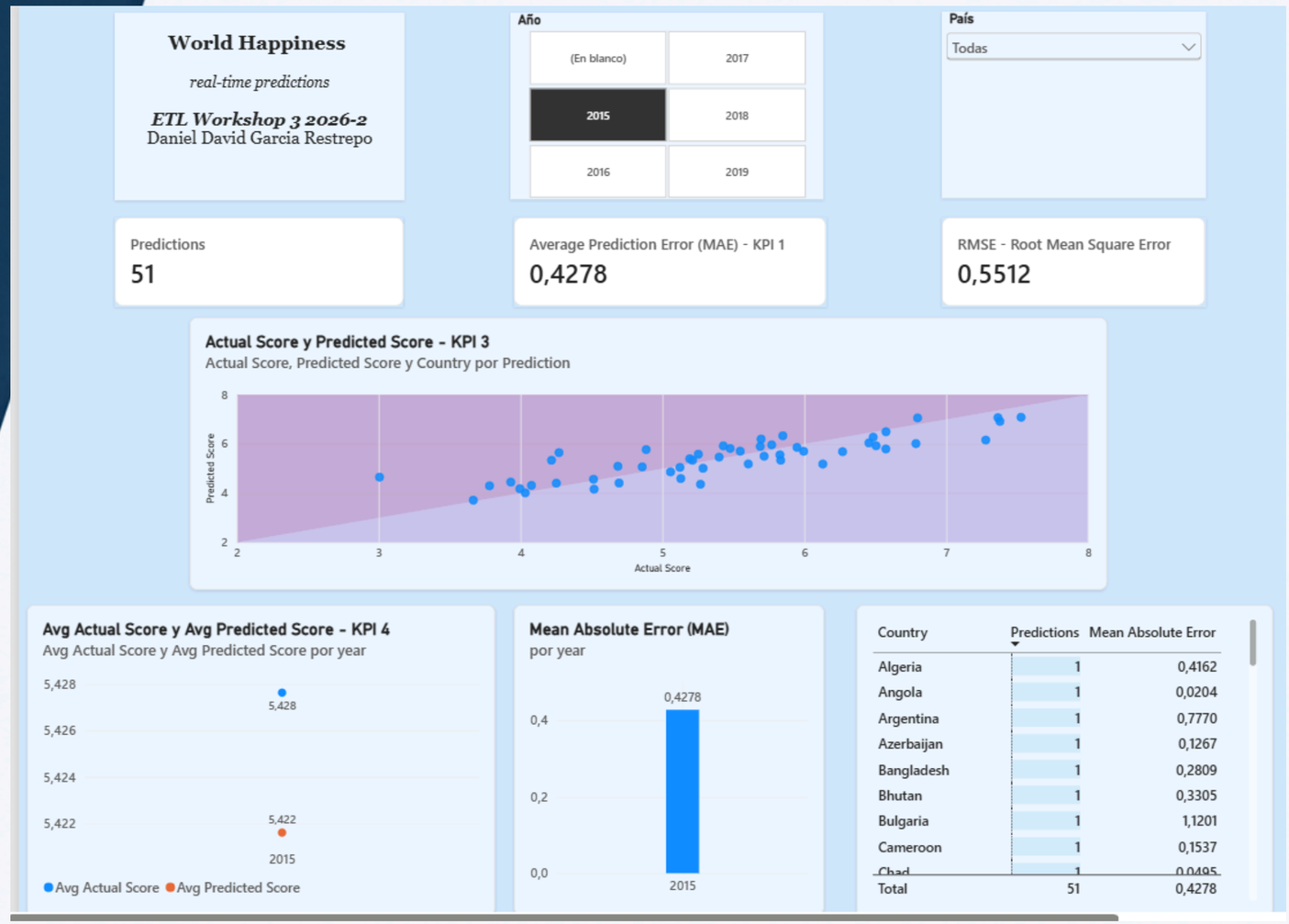
```
 predictions | mean_absolute_error |  rmse
-------------+---------------------+--------
         235 |              0.4321 | 0.5566
(1 row)

      country_name       | predictions | mean_absolute_error
-------------------------+-------------+---------------------
 Albania                 |           4 |              0.5829
 Bangladesh              |           4 |              0.4528
 Tanzania                |           4 |              1.0172
 Turkey                  |           4 |              0.1833
 Algeria                 |           3 |              0.4467
 Argentina               |           3 |              0.4764
 Bahrain                 |           3 |              0.2786
 Burkina Faso            |           3 |              0.2634
 Estonia                 |           3 |              0.5508
 Ghana                   |           3 |              0.1768
 Italy                   |           3 |              0.0595
 Malaysia                |           3 |              0.2113
 Nicaragua               |           3 |              0.4599
 Nigeria                 |           3 |              0.7224
 Northern Cyprus         |           3 |              0.5473
 Qatar                   |           3 |              0.5674
 Russia                  |           3 |              0.2796
 Rwanda                  |           3 |              1.5157
 Singapore               |           3 |              0.7589
 Slovenia                |           3 |              0.4584
```

github.com/danielrestrepo13/ETL-Workshop-3-streaming-ETL-with-Kafka-and-ML

Respaldo

Dashboard filtrado por año (2015)



Trazabilidad

Traceability

event_id	processing_status	country_name	actual_score	predicted_score
E001	INVALID_VALUES			
E001	VALID	Algeria	5,61	5,19
E002	VALID	Angola	4,03	4,01
E003	VALID	Argentina	6,57	5,80
E004	VALID	Azerbaijan	5,21	5,34
E005	VALID	Bangladesh	4,69	4,41
E006	VALID	Bhutan	5,25	5,58
E007	VALID	Bulgaria	4,22	5,34
E008	VALID	Cameroon	4,25	4,41
E009	VALID	Chad	3,67	3,72
E010	VALID	Cyprus	5,69	5,90
E011	VALID	Czech Republic	6,51	5,93